

Wave Release Coordination under Vertical Resource Constraints in Multi-Story AMR Warehouses

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1 Problem Formulation

We consider a multi-story warehouse where floor-bound AMRs share a small number of freight elevators across floors. Orders arrive over time, each with a known source floor and destination floor. A *wave* is a subset of orders released into the system simultaneously at the start of a release window. The system performance metric is the wave makespan, the time at which the last order in the wave is delivered, given a fixed operational dispatch policy.

This setting decomposes naturally into two coupled decision layers. The *tactical* layer chooses which orders to bundle into a wave W , namely the wave composition decision. The *operational* layer determines, given W , which AMR picks which order and which AMRs ride which elevator together. We focus the analytical effort on the tactical layer and treat the operational layer through a fixed dispatch policy (capacity-bounded FIFO boarding; full simulator details appear in Section 5). The two layers are coupled through the makespan: a wave whose composition is structured to ease the downstream dispatch yields a smaller makespan than an unstructured wave of the same size.

Sets and indexes.

- $\mathcal{O}$: set of all orders, indexed by o
- $\mathcal{F} = \{1, 2, \dots, F\}$: set of floors, indexed by f
- $\mathcal{A}$: AMR fleet, indexed by a
- $\mathcal{E}$: set of elevators, indexed by e
- $\mathcal{M} = \{M_1, M_2, M_3\}$: elevator-model set, indexed by m

Parameters.

- $F \in \mathbb{Z}_{\geq 2}$: number of floors
- $|\mathcal{A}|$: AMR fleet size
- E : number of elevators
- $c \in \mathbb{Z}_{\geq 1}$: per-trip elevator capacity
- $s_o \in \mathcal{F}$: source floor of order o
- $d_o \in \mathcal{F}, d_o \neq s_o$: destination floor of order o
- $r_o \in \mathbb{R}_{\geq 0}$: release time of order o
- $W_{\min}, W_{\max} \in \mathbb{Z}_{\geq 1}$: lower and upper bounds on wave cardinality
- $\tau_w \in \mathbb{R}_{\geq 0}$: start time of release window for wave w
- $\Delta \in \mathbb{R}_{> 0}$: length of release window
- $\sigma_{M_3} \in \{0.10, 0.20\}$: lognormal noise scale under stochastic model M_3

Decision variable. The tactical decision variable is the wave composition $W \subseteq \mathcal{O}$, the subset of orders released together within the current wave window. An equivalent binary representation is $x_o \in \{0, 1\}$ with $x_o = 1$ iff $o \in W$.

Structured-feature representation. Rather than working in the raw $|\mathcal{O}|$ -dimensional decision space, we summarize each candidate wave by a three-dimensional fingerprint $\Phi(W) = (C(W), I(W), T(W))$:

- *Vertical concentration* $C(W) = 1 - H_{\text{dst}}(W)/\log_2 F$, where $H_{\text{dst}}(W)$ is the Shannon entropy of the destination-floor distribution within the wave; high C means orders share few destination floors.
- *Directional imbalance* $I(W) = ||W^\uparrow| - |W^\downarrow||/|W|$, where $W^\uparrow$ and $W^\downarrow$ are the up- and down-going subsets; high I means the wave is dominated by a single direction.
- *Temporal clustering* $T(W) = \sigma_r(W)/\mu_r(W)$, the coefficient of variation of per-order release timestamps within the wave; high T means orders arrive in tight bursts.

The three axes correspond to three orthogonal physical mechanisms (multi-floor spread, up-versus-down asymmetry, temporal bunching) by which wave composition affects vertical-resource stress. We treat Φ as a *conceptual* decomposition rather than a predictive surrogate: each axis carries a clear physical unit and operational meaning.

Elevator models. The dispatch policy is realized through a simulator under one of three elevator models:

- M_1 (*throughput aggregation*): each elevator is modeled as a server with a per-AMR throughput rate that absorbs batched-trip efficiency into a single multiplier.
- M_2 (*true co-occupancy batching*): at each elevator, the next c AMRs in the queue board together and ride as a unit, with realistic per-trip capacity, dwell time, and direction handling.
- M_3 (*stochastic batching*): extends M_2 with lognormal noise on per-trip duration at scale σ_{M_3} , used in Section 5 for robustness sensitivity.

M_1 and M_2 are both well-attested in the warehouse OR literature; the *structural* disagreement between throughput aggregation and explicit co-occupancy is the methodological gap our Hedge Rule (Section 4) resolves.

Objective function. Minimize the expected wave makespan under the chosen elevator model:

$$\min_{W \subseteq \mathcal{O}} \mathbb{E}[\mathcal{T}(W) \mid M], \quad (1)$$

where $\mathcal{T}(W) := \max_{o \in W} t_o^{\text{deliver}}(W; M)$ is the wave makespan, $t_o^{\text{deliver}}(W; M)$ is the simulator-realized delivery time of order o under model M , and the expectation is over operational stochasticity (AMR pickup times, elevator dwell times, and, under M_3 , lognormal noise).

Constraints. The wave composition W is subject to the constraints (2)–(6) below: (2)–(5) are tactical-layer feasibility conditions on W , while (6) is operational feasibility realized through the simulator under model M rather than as a formal optimization constraint.

$$W_{\min} \leq |W| \leq W_{\max}, \quad (2)$$

$$r_o \in [\tau_w, \tau_w + \Delta], \quad \forall o \in W, \quad (3)$$

$$\sum_w \mathbf{1}[o \in W_w] = 1, \quad \forall o \in \mathcal{O}, \quad (4)$$

$$x_o \in \{0, 1\}, \quad \forall o \in \mathcal{O}, \quad (5)$$

$$n_{\text{aboard}}(e, t) \leq c, \quad \forall e \in \mathcal{E}, \forall t. \quad (6)$$

Constraint (2) bounds wave cardinality between the warehouse’s operating envelope: $W_{\min}$ prevents under-utilized release windows, and $W_{\max}$ respects picking-station throughput and elevator-queue saturation. Constraint (3) is temporal feasibility: each order’s release time r_o must lie inside the wave’s release window of length Δ . Constraint (4) requires the sequence of waves

to partition $\mathcal{O}$, with no order skipped or double-served, where W_w is the wave released at τ_w and $\mathbf{1}[\cdot]$ is the indicator function. Constraint (5) defines the binary inclusion indicator, equivalently $W = \{o \in \mathcal{O} : x_o = 1\}$. Constraint (6) caps the per-trip elevator load $n_{\text{aboard}}(e, t)$; the simulator additionally encodes three operational invariants not exposed as formal optimization constraints: single-order AMR carriage (one AMR carries at most one order at a time), source-before-destination floor sequencing (an AMR visits s_o before d_o), and capacity-bounded FIFO queue discipline (AMRs board in arrival order, up to c). These invariants are realized through $\mathcal{T}(W)$'s simulator implementation.

Together, equation (1) subject to constraints (2)–(6) defines the *wave release coordination problem under vertical resource constraints*. Direct solution of this stochastic optimization is challenging on two counts: $\mathcal{T}(W)$ is simulator-realized under M rather than closed-form in W , and the decision space (subsets of $\mathcal{O}$) is combinatorial in $|\mathcal{O}|$. Section 4 therefore develops two analytical tools that operate on the structured-feature representation $\Phi(W)$ instead: a Bound-and-Gap decomposition theorem that bounds the value of structured information under any fixed feature partition of Φ space, and a Model-Dominance Hedge Rule that resolves structural disagreement among $\{M_1, M_2, M_3\}$ via a closed-form decision. The four problem-defining elements identified in Sections 1 and 2 (wave composition as decision variable, multi-story deployment, flexible AMR fleet, shared-elevator capacity coupling) are now jointly formalized within this two-stage scheduler.